General Equilibrium

1 Exchange economy

1.1 Basics

1. Partial vs. general equilibrium: the partial equilibrium model discusses the individual market(s), and all prices other than the price(s) of the good(s) being studied are assumed fixed. In the general equilibrium model all prices are variable, and equilibrium requires that all markets clear.

2. We first examine the pure exchange economy, where all agents are consumers. each described by their preferences and the goods (endowments) that they possess. The agents trade the goods among themselves and attempt to make themselves better off.

3. Exogenous variables

    - Consumers $i \in \{1, 2, \ldots, n\}$
    - Goods $j \in \{1, 2, \ldots, k\}$
    - Endowments

    $$\omega = [\omega_1, \ldots, \omega_n] = \begin{bmatrix} \omega_{11} \cdots & \cdots \omega_{1n} \\ \cdot & \cdot \\ \cdot & \cdot \\ \omega_{k1} \cdots & \cdots \omega_{kn} \end{bmatrix}_{k \times n}$$

    - Preferences (or utility functions)

    $$\{\succsim\}_{i=1}^{n} = \{\succsim_1, \succsim_2, \ldots, \succsim_n\}$$

4. Endogenous variables

- Consumption

$$x = [x_1, \ldots, x_n] = \begin{bmatrix} x_{11} \cdots & \cdots x_{1n} \\ \cdot & \cdot \\ \cdot & \cdot \\ x_{k1} \cdots & \cdots x_{kn} \end{bmatrix}_{k \times n}$$

Consumption allocation, x; consumption bundle for agent i, x_i.

- Prices: $p = (p_1, ...p_k)$.

5. Implicit assumptions:

   - private ownership
     Definition: Every portion of every good is owned by exactly one person and that person has the exclusive right to use it in consumption and exchange (or production).
   - No externality, no public goods, no market powers, etc.

1.2 Excess demand and Walras' Law

1. Each consumer i acts as if he or she were solving the following problem

$$\max_{x_i} u_i(x_i); \quad s.t.\ px_i = p\omega_i$$

The answer to this problem, $x_i(p, p\omega_i)$ is the consumer's demand function.

2. The aggregate excess demand function

$$z(p) = \sum_{i=1}^{n} [x_i(p, p\omega_i) - \omega_i]$$

3. Walras' Law. For any price vector p, we have $pz(p) = 0$; i.e., the value of the excess demand is identically zero.

4. Proof: We simply write the definition of aggregate excess demand and multiply by p :

$$pz(p) = p\{\sum_{i=1}^{n} [x_i(p, p\omega_i) - \omega_i]\} = \sum_{i=1}^{n} [px_i(p, p\omega_i) - p\omega_i] = 0$$

since $x_i(p, p\omega_i)$ must satisfy the budget constraint $px_i = p\omega_i$ for each agent $i = 1, ...n$.

5. Market clearing. If demand equals supply in $k-1$ markets, and $p^k > 0$, then demand must equal supply in the kth market. If not, Walras' law would be violated.

1.2.1 Examples

1. Excess demand:

$$z_i(p) = \; x_i(p, \omega_i) - \omega_i$$
$$z(p) = \sum_i \; z_i(p).$$

   Here $x_i(p, \omega_i)$ is the maximal for $\succsim_i$ in $\{ \; x_i | px_i = p\omega_i \}$

2. Example 1.

$$U_A = x_{1A}x_{2A} \qquad \omega_A = (4,1)$$
$$U_B = x_{1B}x_{2B} \qquad \omega_B = (1,4).$$

   In this case,

$$x_{1A}(p_1, p_2, p_1\omega_1 + p_2\omega_2) = \frac{4p_1 + p_2}{2p_1}$$

$$z_{1A}(p) = \tfrac{4p_1+p_2}{2p_1} - 4 = \tfrac{p_2}{2p_1} - 2 = \begin{cases} > 0 & \text{if } p_2 > 4p_1 \\ = 0 & \text{if } p_2 = 4p_1 \\ < 0 & \text{if } p_2 < 4p_1. \end{cases}$$

$$x_{2A}(p_1, p_2, p_1\omega_1 + p_2\omega_2) = \frac{2p_1}{p_2} + \frac{1}{2} \Longrightarrow$$
$$z_{2A}(p) = \frac{2p_1}{p_2} - \frac{1}{2}.$$

$$z_{1B}(p) = \frac{2p_2}{p_1} - \frac{1}{2}$$
$$z_{2B}(p) = \frac{p_1}{2p_2} - 2.$$

$$z(p) = \begin{bmatrix} \frac{5p_2}{2p_1} - \frac{5}{2} \\ \frac{5p_1}{2p_2} - \frac{5}{2} \end{bmatrix}$$

$$p \; z(p) = 0.$$

3. Example 2. Robinson-Crusoe economy

$$U = x^{1/2} + y^{1/2}, \qquad \omega = (1,1).$$

$$z(p) = \begin{bmatrix} \frac{1+p_2/p_1}{1+p_1/p_2} - 1 \\ \frac{1+p_1/p_2}{1+p_2/p_1} - 1 \end{bmatrix} = \begin{bmatrix} \frac{p_2}{p_1} - 1 \\ \frac{p_1}{p_2} - 1 \end{bmatrix}$$

$$p \; z(p) = 0.$$

4. Example 3.

$$u_A = \min\{x_{1A}, x_{2A}\}, \qquad u_B = \min\{x_{1B}, x_{2B}\}$$

$$\omega_A = (4,1), \qquad \omega_B = (1,4).$$

$$z(p) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

5. Example 4.

$$u_A = x_{1A}^{1/2} + x_{2A}^{1/2}, \qquad u_B = x_{1B}.$$

$$\omega_A = (0,1), \qquad \omega_B = (1,0).$$

$$z(p) = \begin{bmatrix} \frac{p_2^2}{p_1p_2+p_1^2} \\ \frac{p_1^2}{p_1p_2+p_1^2} - 1 \end{bmatrix}$$

1.3 Walrasian equilibrium

1. It is natural to think of an equilibrium price vector as being one that clears all markets. We thus define a Walrasian equilibrium to be a pair (p^*, x^*), such that

$$\sum_i x_i(p^*, p^*\omega_i) \leq \sum_i \omega_i$$

   That is, p^* is a Walrasian equilibrium if there is no good for which there is positive excess demand.

2. Edgeworth box.

3. Free goods. If p^* is a Walrasian equilibrium and $z_j(p^*) < 0$, then $p_j^* = 0$. That is, if some good is in excess supply at a Walrasian equilibrium it must be a free good.

4. Desirability. If $p_j = 0$, then $z_j(p) > 0$ for $j = 1, ...k$.

5. Equality of demand and supply. If all goods are desirable and p^* is a Walrasian equilibrium, then $z(p^*) = 0$.

6. Proof. Assume $z_i(p^*) < 0$. Then by the free goods proposition, $p_i^* = 0$. But then by the desirability assumption, $z_i(p^*) > 0$, a contradiction.

1.3.1 Existence of an equilibrium

1. We can normalize prices and express demands in terms of relative prices:

$$p_j = \frac{\widehat{p}_j}{\sum_{j=1}^{k} \widehat{p}_j}$$

the normalized prices must always sum up to 1. Hence we can restrict our attention to price vectors belonging to the $k-1$-dimensional unit simplex:

$$S^{k-1} = \{p \text{ in } R_{+}^{k} : \sum_{i=1}^{k} p_i = 1\}$$

2. Brouwer fixed-point theorem. If $f : S^{k-1} \to S^{k-1}$ is a continuous function from the unit simplex to itself, there is some x in S^{k-1} such that $x = f(x)$.

3. Existence of Walrasin equilibria. If $z : S^{k-1} \to R^k$ is a continuous function that satisfies Walras' law, $pz(p) = 0$, then there exists some p^* in S^{k-1} such that $z(p^*) \leq 0$.

4. Proof. Define a map $g : S^{k-1} \to S^{k-1}$ by

$$g_i(p) = \frac{p_i + \max(0, z_i(p))}{1 + \sum_{j=1}^{k} \max(0, z_j(p))} \quad \text{for } i = 1, ..., k$$

This map is continuous since z and the max function are continuous functions. Furthermore, $g(p)$ is a point in the simplex S^{k-1} since $\sum_i g_i(p) = 1$.

By Brouwer's fixed-point theorem there is a p^* such that $p^* = g(p^*)$; i.e..

$$p_i^* = \frac{p_i^* + \max(0, z_i(p^*))}{1 + \sum_{j=1}^{k} \max(0, z_j(p^*))} \quad \text{for } i = 1, ..., k \tag{1}$$

We will show that p^* is a Walrasian equilibrium. Cross-multiply equation (1) and rearrange to get

$$p_i^* \sum_{j=1}^{k} \max(0, z_j(p^*)) = \max(0, z_i(p^*)) \quad i = 1, ..., k$$

Now multiply each of these k equations by $z_i(p^*)$:

$$z_i(p^*)p_i^*[\sum_{j=1}^{k} \max(0, z_j(p^*))] = z_i(p^*) \max(0, z_i(p^*)) \quad i = 1, ..., k$$

Sum these k equations to get

$$\left[\sum_{j=1}^{k} \max(0, z_j(p^*))\right] \sum_{i=1}^{k} p_i^* z_i(p^*) = \sum_{i=1}^{k} z_i(p^*) \max(0, z_i(p^*))$$

Now $\sum_{i=1}^{k} p_i^* z_i(p^*) = 0$ by Walras' law so we have

$$\sum_{i=1}^{k} z_i(p^*) \max(0, z_i(p^*)) = 0$$

Each term of this sum is greater than or equal to 0 since each term is either 0 or $(z_i(p^*))^2$. But if any term were strictly greater than 0, the equality wouldn't hold. Hence, every term must be equal to 0, which says

$$z_i(p^*) \leq 0 \quad \text{for } i = 1, ..., k.$$

1.3.2 Example: the Cobb-Douglas economy

Let agent 1 have utility function $u_1(x_1^1, x_1^2) = (x_1^1)^\alpha (x_1^2)^{1-\alpha}$ and the endowment $\omega_1 = (1, 0)$. Let agent 2 have utility function $u_2(x_2^1, x_2^2) = (x_2^1)^b (x_2^2)^{1-b}$ and the endowment $\omega_2 = (0, 1)$. Then agent 1's demand function for good 1 is

$$x_1^1(p_1, p_2, I_1) = \frac{\alpha I_1}{p_1}$$

At price (p_1, p_2), income is $I_1 = p_1 \times 1 + p_2 \times 0 = p_1$. Substituting, we have

$$x_1^1(p_1, p_2) = \alpha$$

Similarly, agent 2's demand function for good 1 is

$$x_2^1(p_1, p_2) = \frac{b p_2}{p_1}$$

The equilibrium price is where total demand for each good equals total supply. By Walras' law we only need to find the price where total demand for good 1 equal total supply of good 1:

$$\begin{aligned} x_1^1(p_1, p_2) + x_2^1(p_1, p_2) &= 1 \\ \alpha + \frac{b p_2}{p_1} &= 1 \\ \frac{p_2^*}{p_1^*} &= \frac{1 - \alpha}{b} \end{aligned}$$

Note that, as usual, only relative prices are determined in equilibrium.

1.4 The first theorem of welfare economics

1. Pareto efficiency

   - Definition: x is feasible if $\sum_i x_i \leq \sum_i \omega_i$
   - Definition: A feasible allocation x is (Pareto) efficient if there exists no feasible allocation x' such that $\forall i$

   $$x_i' \succsim_i x_i$$

   and $\exists i,\ x_i' \succ_i x_i$.

2. Walrasian equilibrium

   Definition: An allocation-price pair (x, p) is a Walrasian equilibrium if (1) the allocation is feasible, and (2) each agent is making an optimal choice from his budget set. In equations:

   - $\sum_i x_i = \sum_i \omega_i$.
   - If x_i' is preferred by agent i to x_i, then $px_i' > p\omega_i$.

   This definition allows us to neglect the possibility of free goods.

3. First Theorem of Welfare Economics. If (x, p) is a Walrasian equilibrium, then x is Pareto efficient.

4. Proof. Suppose not, and let x' be a feasible allocation that all agents prefer to x. Then by property 2 of the definition of Walrasian equilibrium, we have

   $$px_i' > p\omega_i \text{ for } i = 1, ...n$$

   Summing over $i = 1, ...n$, and using the fact that x' is feasible, we have

   $$p \sum_i \omega_i = p \sum_i x_i' > \sum_i p\omega_i$$

   which is a contradiction.

1.5 The second welfare theorem

1. Second Welfare Theorem. Consider an exchange economy $(u^i, e^i)_{i \in n}$ with aggregate endowment $\sum_{i=1}^{n} e^i >> 0$, and with each utility function u^i is continuous, strongly increasing, and strictly quasi-concave. Suppose that $\overline{x}$ is Pareto-efficient allocation for $(u^i, e^i)_{i \in n}$, and that endowments are redistributed so that the new endowment vector is $\overline{x}$. Then $\overline{x}$ is a competitive equilibrium allocation of the resulting exchange economy $(u^i, \overline{x}^i)_{i \in n}$.

2. Proof. Because $\overline{x}$ is Pareto efficient, it is feasible. Hence, $\sum_{i=1}^{n} \overline{x}^i = \sum_{i=1}^{n} e^i >> 0$. Consequently, we may apply the existence theorem to conclude that the exchange economy $(u^i, \overline{x}^i)_{i \in n}$ possesses a competitive equilibrium allocation $\widehat{x}$. It only remains to show that $\widehat{x} = \overline{x}$. Now in the competitive economy, each consumer's demand is utility maximization subject to her budget constraint. Consequently, because i demands $\widehat{x}^i$, and has endowment $\overline{x}^i$, we must have
$$u^i(\widehat{x}^i) \geq u^i(\overline{x}^i) \quad \text{for all } i \in n \tag{2}$$
But because $\widehat{x}^i$ is an equilibrium allocation, it must be feasible for the economy. Consequently, $\sum_{i=1}^{n} \widehat{x}^i = \sum_{i=1}^{n} \overline{x}^i = \sum_{i=1}^{n} e^i$, so that $\widehat{x}$ is feasible for the original economy as well. Also by (2) $\widehat{x}$ makes no consumer worse off than the Pareto-efficient allocation $\overline{x}$. Therefore, $\widehat{x}$ cannot make anyone strictly better off; otherwise, $\overline{x}$ cannot be Pareto efficient. Hence, every inequality of (2) must be an equality. To see that now $\widehat{x}^i = \overline{x}^i$ for every i, note that if for some consumer this were not the case, then in the competitive equilibrium of the new economy, that consumer could afford the average of the bundles $\widehat{x}^i$ and $\overline{x}^i$ and strictly increase her utility (by strict quasiconcavity), contradicting the fact that $\widehat{x}^i$ is utility-maximizing in the competitive equilibrium.

3. The "philosophical" importance of these two theorems cannot be exaggerated. First they suggest that the market is an efficient organization (first theorem). This was a perception of Adam Smith which he stated in his famous metaphor of "the invisible hand." What is more surprising, the results also show that the market permits the attainment of any efficient allocation (second theorem). So even those among us who find revenue distribution to be uneven can turn to the market to realize an optimum once the desired redistribution has been carried out. In principle, this is the position adopted by the proponents of "market socialism," and among them Leon Walras himself can be counted.

4. Note that the hypothesis behind these two theorems merit examination. And we will discuss them in details later. Namely, they assume the existence of a complete set of markets, no public goods, no externalities, the convexity, and the pure and perfect competition, complete information.

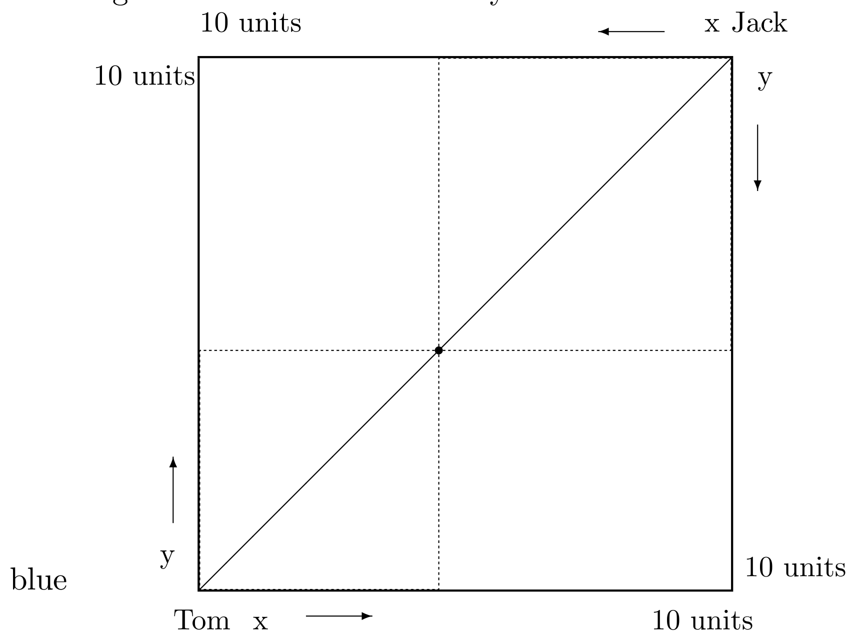
1.6 Simple examples

1. Two typical consumers: Tom has 10 units of goods x, and Jack has 10 units of y. They both have same utility function

$$U_T(x_T, y_T) = (x_T y_T)^{1/2} \qquad U_J(x_J, y_J) = (x_J y_J)^{1/2}.$$

2. Edgeworth box: An Edgeworth box shows all the possible allocations of goods that are possible in the economy, given the total supply of goods.

   The Edgeworth box in this economy is



3. Feasible allocation: any points in the Edgeworth box such that

$$x_T + x_J \leq 10,$$

$$y_T + y_J \leq 10.$$

4. Pareto efficient allocation: An allocation is Pareto efficient if there is no feasible allocation that can make one agent better off without hurting others.

   Pareto efficient allocation in exchange economy

$$MRS^T_{x,y} = MRS^J_{x,y}.$$

   The Pareto efficient allocation in this exchange economy is the diagonal line

$$x_T = y_T, \qquad x_T + x_J = 10 \qquad y_T + y_J = 10.$$

5. Contract curve gives all efficient allocation in the Edgeworth box.

6. Competitive equilibrium is a price vector (p_x, p_y) and the corresponding allocations such that given (p_x, p_y), (x_T, y_T) maximizes Tom's utility and (x_J, y_J) maximizes Jack's utility, and markets for both goods are cleared.

   Solving for the competitive equilibrium

   Given utility function, we solve the utility maximizing problem for Tom and Jack,

   $$x_T = \frac{I_T}{2p_x},\ y_T = \frac{I_T}{2p_y} \text{ where } I_T = 10p_x.$$

   $$x_J = \frac{I_J}{2p_x},\ y_J = \frac{I_J}{2p_y} \text{ where } I_J = 10p_y.$$

   Plugging I_T, I_J into the allocations yields

   $$x_T = 5,\ y_T = \frac{5p_x}{p_y},\ x_J = \frac{5p_y}{p_x},\ y_J = 5.$$

   In equilibrium,

   $$5 + \frac{5p_y}{p_x} = 10,$$
   $$5 + \frac{5p_x}{p_y} = 10.$$

   Therefore we have the competitive equilibrium (let $p_x = 1$)

   $$p_x = 1,\ p_y = 1,\ , x_T = y_T = 5,\ x_J = y_J = 5.$$

   The equilibrium allocation is Pareto efficient.

2 Production

2.1 Firms

1. Suppose there are J firms. Let $y_j \in R^n$ be a feasible production plan for firm j. y_j has negative entries for net inputs and positive entries indicating net outputs. For example, $y_j = (-6, 2)$, then the production plan requires 6 units of commodity one as an input, to produce 2 units of commodity two as an output.

2. The set of feasible production plans, the production possibility set, for firm j is Y_j. A production plan y_j is efficient if there is no y'_j in Y_j such that $y'_j > y_j$. The graph for the set of efficient production plans is the production possibility frontier (PPF). And the slope of the PPF is the marginal rate of transformation (MRT).

3. Markets are competitive and firms are price-taking. Each firm faces fixed commodity prices $p \geq 0$ and chooses a production plan to maximize profit. That is

$$\max_{y_j \in Y_j} p y_j$$

   Because of our sign convention inputs are accounted for in profits as costs and output as revenues.

4. The net supply function $y_j(p)$ of a competitive firm is simply the function that associates to each vector p the profit-maximizing net output vector at those prices.

5. If we have m firms, the aggregate net supply function will be $y(p) = \sum_{j=1}^{m} y_j(p)$.

6. An aggregate production plan, y, maximizes aggregate profit iff each firm's production plan y_j maximizes its individual profit.

2.2 Consumers

1. Distribution of profits. Consumers own firms and are entitled to a share of the profits. We summarize this ownership relation by a set of numbers θ_{ij}, where θ_{ij} represents consumer i's share of the profits of firm j. For any firm j we require that $\sum_{i=1}^{n} \theta_{ij} = 1$.

2. The budget constraint of the consumer now becomes

$$p x_i = p \omega_i + \sum_{j=1}^{m} \theta_{ij} p y_j(p)$$

3. The consumer will choose a utility-maximizing bundle that satisfies this budget constraint. We can get the consumer i's demand function $x_i(p)$.

4. Adding together all of the consumers' demand function gives the aggregate consumer demand function $x(p) = \sum_{i=1}^{n} x_i(p)$.

2.3 Walras' Law

1. The aggregate supply vector comes from adding together the aggregate supply from consumers, which we denote by $\omega = \sum_{i=1}^{n} \omega_i$, and the aggregate net supply of firms, $y(p)$.

2. The aggregate excess demand function is

$$z(p) = x(p) - y(p) - \omega$$

3. Walras' law. $pz(p) = 0$ for all p.

4. Proof.

$$\begin{aligned} pz(p) &= p[x(p) - y(p) - \omega] \\ &= p\left[\sum_{i=1}^{n} x_i(p) - \sum_{j=1}^{m} y_i(p) - \sum_{i=1}^{n} \omega_i\right] \\ &= \sum_{i=1}^{n} px_i(p) - \sum_{j=1}^{m} py_i(p) - \sum_{i=1}^{n} p\omega_i \end{aligned}$$

The budget constraint of the consumer is $px_i = p\omega_i + \sum_{j=1}^{m} \theta_{ij} py_j(p)$.

Making the replacement:

$$\begin{aligned} pz(p) &= \sum_{i=1}^{n} p\omega_i + \sum_{i=1}^{n}\sum_{j=1}^{m} \theta_{ij} py_j(p) - \sum_{j=1}^{m} py_i(p) - \sum_{i=1}^{n} p\omega_i \\ &= \sum_{j=1}^{m} py_j(p) \sum_{i=1}^{n} \theta_{ij} - \sum_{j=1}^{m} py_i(p) = 0 \end{aligned}$$

2.4 First Theorem of Welfare Economics

1. An allocation (x, y) is feasible if aggregate demand are compatible with the aggregate supply

$$\sum_{i=1}^{n} x_i - \sum_{j=1}^{m} y_j - \sum_{i=1}^{n} \omega_i = 0$$

2. A feasible allocation (x, y) is Pareto efficient if there is no other feasible allocation (x', y') such that $x'_i \succ x_i$ for all $i = 1, ..., n$.

3. The first theorem of welfare economics. If (x, y, p) is a Walrasian equilibrium, then (x, y) is Pareto efficient.

4. Proof. Suppose not, and let (x', y') be a Pareto dominating allocation. Then since the consumers are maximizing utility we must have

$$px'_i > p\omega_i + \sum_{j=1}^{m} \theta_{ij} py_j \text{ for all } i = 1, ..., n$$

Summing over the consumer $i = 1, ..., n$, we have

$$p \sum_{i=1}^{n} x_i' > \sum_{i=1}^{n} p\omega_i + \sum_{j=1}^{m} py_j$$

Now we use the feasibility of x' and replace $\sum_{i=1}^{n} x_i'$ by $\sum_{i=1}^{n} \omega_i + \sum_{j=1}^{m} y_j'$:

$$p\left[\sum_{i=1}^{n} \omega_i + \sum_{j=1}^{m} y_j'\right] > \sum_{i=1}^{n} p\omega_i + \sum_{j=1}^{m} py_j$$
$$\sum_{j=1}^{m} py_j' > \sum_{j=1}^{m} py_j$$

But this says that aggregate profits for the production plans y_j' are greater than y_j, which contradicts profit maximization by firms.

2.5 Second theorem of welfare economics

1. The second theorem of welfare economics. Suppose (x^*, y^*) is a Pareto efficient allocation in which each consumer holds strictly positive amounts of each good, and the preferences are convex, continuous, and strongly monotonic. Suppose firms' production possibility sets, y_j, for $j = 1, .., m$ are convex. Then there exists a price vector $p \geq 0$ such that (1) if $x_i' \succ x_i^*$, then $px_i' > px_i^*$ for $i = 1, ...n$; (2) if y_j' is in Y_j, then $py_j^* \geq py_j'$ for all y_j' in Y_j, for $j = 1, ..., m$.

2. The above proposition shows that every Pareto efficient allocation can be achieved by a suitable reallocation of "wealth." We determine the allocation (x^*, y^*)that we want, then determine the relevant prices p. If we give consumer i income px^*, she or he will not want to change her or his consumption bundles.

3. In other words, the second welfare theorem shows that under some assumptions, every Pareto optimum can be decentralized as a competitive equilibrium of a private property economy where resources have been redistributed through lump-sum transfers. When this theorem applies, the government can choose its preferred Pareto optimum, proceed to the right lump-sum transfers, and let the competitive equilibrium work its magic without any other kind of intervention.

4. In practice, we observe several phenomena that lead to market failure. First come public goods: these are by definition the nonrival goods, which one agent can consume without reducing the consumption of other agents. Then the second theorem doesn't apply any more (technically, consumption do not add across agents). The optimal production level of the public good cannot be attained without an intervention of government. Moreover these public goods must be financed, which raises the classical free-rider problem.

5. The presence of externality also implies that the market cannot reach an optimum on its own, corrective taxes are one way to remedy this market failure (as with tax designed to reduce pollution). Adam Smith already considered that the prince must provide three categories of public goods to his subjects: defense, justice, and public work, plus a private good subject to externalities: primary education. On the other hand, he thought that higher education must be left to the private sector, with teachers paid on a performance basis.

6. Even in the hypothetical absence of these market failures, lump-sum transfers are very unlikely to be a practical proposition. Computing the optimal lump-sum transfers requires government to have an extraordinarily detailed information on the characteristics of the economy.

2.6 Examples

2.6.1 The Cobb-Douglas constant returns economy

One consumer has a Cobb-Douglas utility function for consumption, x, and leisure, $h : u(x,h) = a\ln x + (1-a)\ln h$. The consumer is endowed with one unit of labor/leisure and there is one firm with a CRS technology: $x = aL$.

We see the equilibrium real wage must be the marginal product of labor, hence, $w^*/p^* = a$. The maximization problem of the consumer is

$$\max a\ln x + (1-a)\ln h; \quad \text{s.t. } px + wh = w$$

in writing the budget constraint, we have used the fact that equilibrium profits are zero. We find that

$$x(p,w) = a\frac{w}{p}$$
$$h(p,w) = 1-a$$

Hence, equilibrium supply of labor is a, and the equilibrium output is a^2.

2.6.2 A decreasing return to scale economy

Suppose the consumer has a Cobb-Douglas utility function as in the last example, but the producer has a production function $x = \sqrt{L}$. We arbitrarily normalize the price of output to be 1. The profit maximization problem is

$$\max \ \sqrt{L} - wL$$

By the first-order condition we have

$$\frac{1}{2}L^{-\frac{1}{2}} - w = 0$$

Solving for the firm demand and supply functions:

$$L = (2w)^{-2}$$
$$x = (2w)^{-1}$$

The profit function is

$$\pi(w) = (4w)^{-1}$$

The income of the consumer now includes the profit income so that the demand for leisure is

$$h(w) = \frac{(1-a)}{w}(w + \frac{1}{4w}) = (1-a)(1 + \frac{1}{4w^2})$$

By Walras' law we only need find a real wage that clears the labor market:

$$\frac{1}{4w^2} = 1 - (1-a)(1 + \frac{1}{4w^2})$$

Solving this equation, we find

$$w^* = \left(\frac{2-a}{4a}\right)^{\frac{1}{2}}$$

The equilibrium level of profits is therefore

$$\pi^* = \frac{1}{4}\left(\frac{2-a}{4a}\right)^{\frac{1}{2}}.$$

2.6.3 Homework

1. (J&R), 3.48; 3.51; 4.9; 4.13; 4.22; 4.26; 5.12; 5.18; 5.32.

2. Due: Dec. 18, hard copy please.